Product Name					Date completed						Additional Comments				
Stage					Team members										
Sr.	Process step/System function	Failure mode	Effect of the failure	Severity	Root cause of the failure	Occurrence	Controls	Detection	RPN	Recommended actions	Action owner and Target date	New Severity	New Occurrence	New Detection	New RPN
	What this step or function does?	What can possibly go wrong?	What happens if it goes wrong?	How bad would it be? (1-10) How bad would it be? (1-10)	Why would it go wrong?	How often will it go wrong? (1-10) How often will it go wrong? (1-10)	What controls are in place already to prevent it?	How quickly can we detect if failure occures? (10-1) How quickly can we detect if failure occures? (10-1)	S x O x D	What is recommended to prevent occurrence of root cause or reduce severity or early detection of failure?	Who will take action on recommendations and by when?	Depending upon actions taken, what is new level of severity? (1-10) Depending upon actions taken, what is new level of severity? (1-10)	Depending upon actions taken, what is new level of occurrence? (1-10) Depending upon actions taken, what is new level of occurrence? (1-10)	Depending upon actions taken, what is new level of detection? (10-1) Depending upon actions taken, what is new level of detection? (10-1)	New S x New O x New D

Pre-mortem analysis

is from zero to ten; zero being the least severe, while ten is the most severe consequence(s).

Root cause. For each failure listed in column #2, root cause analysis is done to find an answer to the question—What will cause this step of function to go wrong?

Occurrence. This column is another rating based on the frequency of failure. How frequently are these failures, as listed in column #2, likely to occur? Occurrence is ranked on a scale of one to ten, where one is a low occurrence, and ten is a high or frequent occurrence.

Controls. An answer to the question—What controls are in place currently to prevent potential failure as per column #2? What controls are in place to detect the occurrence of a fault, if any?

Detection. This is another rating column where ease of detection of each failure is assessed. Typical questions to ask are: How easy is it to detect each of the potential failures? What is the likelihood that you can discover these potential failures promptly or before they reach the customers? Detection is ranked on a scale of ten to one (please note reversal of the scale). Here rating of one means easily and quickly detectable failure, whereas ten means unlikely and very late detection of failure. Late detection often means a more problematic situation, and therefore the rating for late detection is higher.

RPN (Risk Priority Number). The risk priority is determined by multiplying all the three ratings from column #4, 6, and 8. So, RPN = Severity x occurrence x detection. Thus, a high RPN would indicate a high-risk process step or solution function (as in column #1). Accordingly, steps or functions with higher RPN warrant immediate attention for fixing.

Recommended actions. In this column, SMEs would recommend one or more actions to handle the risks identified. These actions may be directed towards reducing the severity, chances of failure occurrence, improving the detection level, or maybe all of the above.

Action owner and target date. This column is essential from the project management point of view as well as for tracking. Each recommended action can be assigned to a specific owner and carried out before the target date to contain the risks.

Actions completed. This column lists all the actions taken, recommended, or otherwise to lower the risk level (RPN) to an acceptable level or lower.

New severity. Once the actions listed in column #12 are complete, the same exercise must be repeated to arrive at a new level of severity.

New occurrence. The occurrence must have changed depending on actions completed, so this column has a new occurrence rating.

New detection. Due to risk mitigation actions, detection must have changed, too. Register it in this column.

New RPN. Due to change in severity, occurrence, and detection ratings, risk level would have changed. A new RPN is calculated in the same way (severity x occurrence x detection) and recorded in this column.

A usable and handy template for pre-mortem analysis is available for free download at *https://www.anandtamboli.com/tools#pre-mortem*

More about ratings

Several risk analysis methodologies often recommend only two rating evaluations, i.e., severity and occurrence. However, in the case of pre-mortem analysis, we are using the third rating—Detection.

Early detection of the problem can often enable you to contain the risks before becoming significant and out of control. This way, you can either fix the system immediately or may invoke systemwide control measures to remain more alert. Either way, being able to detect failures quickly and efficiently is an advantage in complex systems like AI.

In case of severity and occurrence ratings, the scale of one to ten does not change—no matter what type of solution or industry you are doing it for.

In implementing pre-mortem analysis, you must take a pragmatic approach and choose the scale as